Uncovering LLM Lineage via Static Fingerprints: Progress, Gaps, and Opportunities

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Abstract

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Do, Not, Use, This, Code, Put, the, Correct, Terms, for, Your, Paper

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1 Introduction

Large Language Models (LLMs) have significantly transformed the landscape of artificial intelligence and software engineering. While proprietary models such as GPT and Claude dominate the consumer market, public open-source models such as DeepSeek are increasingly gaining popularity among business entities due to their enhanced transparency, lower deployment costs, reduced legal and licensing constraints, and greater flexibility for customization. Indeed, popular model-sharing platforms (e.g., Hugging Face) host over 2 million models [2], with top models receiving more than one million downloads [45].

Notably, a closer examination of the interconnections among these models reveals a significant degree of reuse [45]. In particular, many models are derived from existing base models through techniques such as fine-tuning [22], model merging [6], and quantization [94]. Such derivations enable the adaptation of general-purpose models to domain-specific scenarios, capability enhancement, and efficiency optimization. However, while enjoying the benefits, the model reusing also bring the inherited risks [13, 18, 32, 79], including legal liabilities (e.g., Intellectual Property violations) to security threats [25, 42, 44] (e.g., backdoors, jailbreaks, and prompt injection).

An effective strategy for risk mitigation is to ensure transparency in model composition, such as using AI Bills of Materials (AI-BOM) [1]. However, although platforms provide mechanisms (e.g., model cards in Hugging Face [5]) that allow contributors to document meta-information, such as model reuse details or model performance [60], such practices are not consistently adopted proactively by model developers. This leads to incomplete disclosure of model information. For instance, prior work reports that the ninth most downloaded pretrained model on Hugging Face, with over 6.9 million monthly downloads, lacks a model card [45]. Consequently, it undermines model transparency, thereby hindering effective risk traceability and mitigation.

To address this problem, a range of model fingerprinting techniques can be leveraged. While dynamic fingerprinting approaches rely on analyzing a model's inference behavior in response to carefully designed input queries [26, 27, 63, 65, 66, 87–91, 95, 107], they are often sensitive to input selection and may suffer from limited robustness under model modifications such as fine-tuning or compression. Therefore, we focus on the problem of *model lineage construction via static fingerprints*.

Empirical Study. We begin with an empirical study to understand the dependency prevalence and dependency types of public model reuse.

The problem of opaque model lineage is increasingly pressing. LLMs are often trained on massive datasets collected from multiple sources, combined with pre-trained models, and further fine-tuned in proprietary pipelines. This complexity obscures the relationship between models, making it difficult to trace the origin of errors, detect unintended biases, or comply with licensing requirements. Consequently, software engineers, AI practitioners, and policymakers face challenges in maintaining trust, reliability, and accountability in AI-driven systems. Despite the growing literature on supply chain analysis in traditional software and deep learning frameworks, systematic studies focused on LLM lineage remain sparse.

This paper seeks to address this gap by investigating how far current methods go in tracing and analyzing the lineage of LLMs. Specifically, we ask: (1) What techniques exist to track dependencies and derivations among LLMs? (2) How effective are these techniques in providing visibility into model origins and transformations? (3) What limitations persist, and where is further research needed? By framing LLM lineage in the context of software engineering, we aim to connect advances in AI with established practices in software maintenance, traceability, and quality assurance.

2 Motivation

The proliferation of Large Language Models (LLMs) has been dramatically accelerated by collaborative platforms like the Hugging Face Hub, which serves as a central repository for model sharing. A cornerstone of this ecosystem is the iterative development of new models by building upon existing ones through techniques such as fine-tuning, merging, and quantization. This process creates a complex web of inter-model dependencies, which we define as **model genealogy** or **lineage**. A clear understanding of a model's genealogy is paramount for ensuring traceability, reproducibility,

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and compliance, as derivative models often inherit biases, vulnerabilities, or licensing constraints from their predecessors.

However, the mechanisms currently available on the Hugging Face Hub for documenting this critical lineage information are frequently inconsistent, incomplete, or entirely absent. This lack of a systematic and enforceable standard for provenance documentation poses a significant barrier to researchers and developers. The most visible symptom of this issue is the difficulty, or outright impossibility, of programmatically constructing a model's lineage–a structure ideally represented as a "Model Tree". This is not a superficial interface problem but a direct consequence of fundamental flaws in how provenance data is recorded at its source.

To illustrate the severity of this challenge, we will dissect two distinct modes of documentation failure, each representing a progressively worsening degree of information obfuscation.

2.1 Problem 1: Lineage Concealed in Unstructured Text

In the most common scenario, lineage information exists but is locked within free-form text, rendering it inaccessible to automated tools. This forces researchers into a manual, error-prone process of discovery.

Case Study 1: `DavidAU/NSFW_DPO_Noromaid-7b-GGUF`

The Hugging Face repository of this model does not include a model tree; however, its model card explicitly states:

> *NSFW_DPO_Noromaid-7b-Mistral-7B-Instruct-v0.1 is a merge of the following models: mistralai/Mistral-7B-Instruct-v0.1 and athirdpath/NSFW_DPO_Noromaid-7b.*

Notably, the Hugging Face pages of both referenced models in the model card contain their respective model trees. This reliance on unstructured documentation makes the automated construction of a lineage graph infeasible. It imposes a significant manual burden for identifying even a direct parent and completely obstructs any large-scale, systematic analysis of the model ecosystem.

Case Study 2: `timdettmers/guanaco-13b`

Another example is the `timdettmers/guanaco-13b` model. While this model is a fine-tuned version of LLaMA, this pivotal relationship is not captured in any structured, machine-readable metadata. To discover this link, a user must manually navigate to the model's repository, open the "Model Card" (a Markdown file), and parse through natural language prose to find the statement:

> *Guanaco models are open-source finetuned chatbots obtained through 4-bit QLoRA tuning of LLaMA base models...*

2.2 Problem 2: Lineage Outsourced to Fragile External Links

A more precarious situation arises when provenance information is not even located within the Hugging Face platform, but is instead hyperlinked to an external, third-party source.

Case Study 3: `pfnet/plamo-2.1-2b-cpt`

The model's homepage lacks a model tree, and its model card merely states that the model was derived by reducing the parameter size from an 8B-parameter model, without explicitly specifying the base model used for this operation. However, the model card provides links to three technical blogs for reference. By following these external links, we can infer that the mentioned 8B-parameter model is `plano-2-8b`.

Case Study 4: `Remilistrasza/Mysterious_SDXL_Lah`

Consider the `Remilistrasza/Mysterious_SDXL_Lah` model. Its Model Card is starkly empty of descriptive content, offering only a single hyperlink to a page on Civitai, another model-sharing platform. To identify the base model, a user is forced to leave the Hugging Face ecosystem. Only upon navigating to this external page does one discover that the base model is "*SDXL 1.0*".

This practice introduces two severe risks: **information fragmentation** and **link rot**. The integrity and availability of the lineage information become wholly dependent on the uptime and archival policies of an external website. Should the link become invalid (a 404 error), or the content on the remote page be altered or deleted, the model's traceable lineage is permanently severed. This practice renders the model's provenance fragile and unverifiable from within the Hugging Face ecosystem.

These documented failures, which range from information concealed in prose to its complete omission, demonstrate a clear and pressing need for an automated and reliable method to construct model genealogies. The current reliance on inconsistent and manual documentation is an unsustainable bottleneck in an ecosystem that is scaling rapidly. To bridge this critical gap, we propose an approach to automatically identify and verify the derivative relationships between models, thereby enabling robust and scalable lineage tracing.

2.3 Literature on Model Lineage Construction

As the problem of model traceability gains prominence, various methods have been proposed to detect relationships between LLMs (summarized in Table 1). However, these approaches typically over-optimize for specific operational scenarios, severely limiting their generalizability when constructing comprehensive lineage trees in the wild. First, the majority of techniques (e.g., REEF [107], MDIR [108], HuRef [105]) treat provenance as a symmetrical, pairwise similarity problem (Type "P") rather than a directional graph, failing to identify which model is the parent and which is the child (Dir. "×"). Second, they suffer from strict evaluation biases: dynamic

Name	Year	Acc.	Analy.	Type	Dir.	Evolution Support
TensorGuard [88]	2025	W	D	P	×	FT, PT
LLM DNA [87]	2026	B	D	L	×	FT, PT, D
Model Map [63]	2025	W	D	L	✓	FT, M, Q, D
REEF [107]	2025	W	D	P	×	FT, M, P, D
MDIR [108]	2025	W	S	P	×	FT, P, UC
PhyloLM [98]	2025	B	D	L	×	FT, Q
Mother [30]	2025	W	S	L	✓	FT, PT
Model Atlas [29]	2025	W	S	L	✓	FT, M, Q, D
Neural Lineage [101]	2024	W	S, D	P	×	FT
HuRef [105]	2024	W	S	P	×	FT, PT
SeedPrints [76]	2026	W	D	P	×	FT, M, PT, Q, D
PDF [100]	2025	W	S	P	×	FT, UC

Table 1: Acc.(ess): Black/White-box; Analy.(sis): Static, Dynamic; Type: Pair-wise, Lineage-tree; Dir.: Directional inference; Evolution: FT: Fine-tuning, M: Merging, PT: PEFT, Q: Quantization, P: Pruning, UC: Upcycling, D: Distillation.

methods rely on specific test prompts that are vulnerable to output homogeneity [63, 87, 88, 98, 107], while static weight-based methods often require rigid architectural alignment [100, 105, 108]. Consequently, while highly accurate for standard fine-tuning (FT), most methods fail to generalize across the complex, combinatorial operations actually prevalent, such as model merging (M), quantization (Q), Distillation (D) and pruning (P). Although a few recent works like Mother [30] and Model Atlas [29] attempt directional lineage tracing, they still suffer in real-world scenarios due to limited evolution support and fragile directional inference: Mother is strictly confined to basic fine-tuning, while Model Atlas relies on intra-architecture weight distances that completely fail to capture cross-architecture evolutions (e.g., distillation) or subtle alignments (e.g., RLHF). This highlights the urgent need for a more generalized, operation-agnostic framework for automated genealogy construction.

2.4 A Preliminary Investigation on Hugging Face Model Card

[29]

3 Benchmark

3.1 Construction

The construction of our benchmark involves a systematic pipeline that traces model derivations from base models to their descendants on Hugging Face. This process is divided into five key steps:

Step 1: Obtaining Base Model List. We begin by collecting a list of popular base models for the `text-generation` task from Hugging Face. Utilizing the `HfApi` provided by `huggingface_hub`, we retrieve the top 50 non-gated models sorted by the number of likes. The model identifiers are stored in a file named `base_models.txt`, where each line corresponds to one base model ID. This step is implemented via **Script 1**.

Step 2: Building the Model Derivation Tree. Based on `base_models.txt`, we recursively crawl derivative models to construct a "model forest". For each base model, we explore its derivatives up to four levels deep, covering four derivation types: `finetune`, `adapter`, `quantized`, and `merge`. For each derivation type, a maximum of six sibling nodes are considered. Every node in the tree records metadata such as model name, crawl timestamp, depth level, and its child nodes. The resulting forest structure is serialized into `model_forest.json`. This step is implemented via **Script 2**.

Step 3: Constructing Model Pairs. From the model forest, we extract two categories of model pairs: adjacent parent-child pairs and non-adjacent ancestor-descendant pairs. By performing a Breadth-First Search (BFS) traversal on each tree, we generate:

- **Adjacent Pairs**: Direct parent-child model pairs.
- **Non-Adjacent Pairs**: All ancestor-descendant pairs where the ancestor is not the immediate parent.

These pairs are recorded in `adjacent_pairs.jsonl` and `non_adjacent_pairs.jsonl`, respectively. This step is implemented via **Script 3**.

Step 4: Single-Threaded Model Validation. To ensure data quality, we validate the structural correctness of all adjacent model pairs. For each model, we inspect its file tree on Hugging Face, checking for the presence of critical files such as `config.json`, `adapter_config.json`, and various parameter files (e.g., `.bin`, `.pt`, `.safetensors`, etc.). We verify whether the claimed derivation type aligns with the actual file structure and whether the parent model is valid. The results are compiled into a comprehensive validation report named `model_validation_report.json`. This step is implemented via **Script 4**.

Step 5: Filtering Erroneous Model Pairs. Based on the validation report, we remove all erroneous model pairs from the dataset. Models flagged with any inconsistencies or missing critical files are excluded. The cleaned set of adjacent model pairs is saved in `adjacent_pairs_without_error.json`. This step is implemented via **Script 5**.

3.2 Characteristics

Our benchmark exhibits several notable characteristics that reflect its diversity and representativeness of the Hugging Face model ecosystem:

- **Variety of Derivation Types**: The dataset covers four major derivation types—`finetune`, `adapter`, `quantized`, and `merge`—capturing a wide range of model transformation patterns.
- **Hierarchical Depth**: With a recursive crawling depth of up to four levels, the benchmark contains both shallow and deep derivation chains, enabling analysis of both immediate and long-range model inheritance.
- **Scale and Coverage**: Starting from 50 popular base models, the recursive traversal results in a substantial model forest, ensuring coverage of highly influential models as well as their derivative variants.
- **Data Cleanliness**: Through rigorous validation and error filtering, the dataset ensures high structural integrity, making it suitable for tasks that require reliable model lineage information.
- **Real-World Relevance**: The benchmark reflects actual user-driven derivation patterns on Hugging Face, providing insights into community practices such as model fine-tuning trends, adapter usage, and quantization adoption.

Table 2: Statistics of the GrandMoTher benchmark.

Statistic	Value
#Graphs	31
#Model pairs (edges)	248
Min / Median / Max pairs per graph	1 / 6 / 22
Avg. pairs per graph	8.0
#Models (nodes)	279
#Root models	31
#Intermediate models	44
#Leaf models	204
Share of leaf models	73.1%
Min / Avg / Max models per graph	2 / 9.0 / 23
Avg. out-degree of non-leaf models	3.31
Max fine-tuning depth	2

4 Empirical Setup

We design experiments to explore the following research questions.

- **Measurement Evaluation (RQ1)**: How do different measurement metrics affect the effectiveness of LLM lineage detection?
- **Layer Contribution Analysis (RQ2)**: How do different model layers influence the effectiveness of LLM lineage detection?
- **Failure Analysis (RQ3)**: Why do existing methods fail to detect lineage?
- **Integration (RQ4)**: using the above findings, the best performance can achieve? (80%)

Measurement Metrics. We select common measurement metrics that are used in static white-box fingerprinting. Static white-box fingerprinting captures intrinsic weight characteristics, including statistical distributions [100] (e.g., L_2 distances of model tensors [30]), directionally invariant terms [105], SVD-based spectral signatures [78], and matrix-driven homology proofs [108].

Accuracy. To comprehensively evaluate the structural recovery of model lineages, we measure performance at both the edge level and the tree level. Let $\mathcal{T}$ denote the set of all model family trees in our dataset. We define the following two metrics:

- *Micro-accuracy*: This metric evaluates the local edge prediction performance. It is defined as the total number of correctly predicted directed edges divided by the total number of ground-truth edges across all evaluated model graphs:

$$Micro\text{-}acc = \frac{\sum_{T \in \mathcal{T}} |E^{(T)}_{correct}|}{\sum_{T \in \mathcal{T}} |E^{(T)}_{total}|}$$

- *Macro-accuracy*: This metric evaluates the global topology reconstruction. It is defined as the ratio of successfully recovered model family trees (where the entire lineage topology and edge directions are perfectly matched) to the total number of model families:

$$Macro\text{-}acc = \frac{|\{T \in \mathcal{T} \mid T_{recovered} = T_{true}\}|}{|\mathcal{T}|}$$

Layers. To address RQ2, we systematically isolate the core structural components within Transformer-based Large Language Models (LLMs). Specifically, we dissect the Multi-Head Attention mechanism and the Feed-Forward Network (FFN). The attention mechanism involves projection weight matrices for the Query (W_Q), Key (W_K), Value (W_V), and Output (W_O):

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

Furthermore, we evaluate the contribution of the FFN, which typically consists of Multi-Layer Perceptron (MLP) dense layers:

$$FFN(x) = \sigma(xW_1 + b_1)W_2 + b_2$$

where σ represents a non-linear activation function. This granular breakdown enables us to systematically evaluate the fingerprinting capacity of various layer combinations and quantify their respective impacts on the effectiveness of lineage detection.

Graph Algorithms. In our experiments, we evaluate and compare several graph construction algorithms to reconstruct the model lineage topology from the computed pairwise distance and directional matrices:

- *MoTher*: Following Horwitz et al. [30], we formulate the base lineage recovery task as finding a minimum directed spanning tree (MDST). We employ the Chu-Liu-Edmonds algorithm [14, 17] to achieve this optimal cycle-free directed routing.
- *Atlas Greedy DAG*: To account for complex, non-tree structures, we evaluate the Model Atlas framework [29]. This approach constructs a Directed Acyclic Graph (DAG) via a greedy algorithm, leveraging real-world structural priors such as temporal upload dynamics to determine edge directionality and explicitly accommodating multi-parent merged models.
- *Strict-Directional MDST*: We introduce a heavily constrained variant of the MDST. By setting the penalty for reverse edges (i.e., edges that violate the inferred directionality constraints) to infinity (∞), we strictly force the algorithm to adhere to the predicted parent-child hierarchical flow.
- *Undirected-MST*: We also implement a decoupled baseline. In the first stage, we construct a minimum spanning tree (MST) on the undirected graph using solely the distance matrix to determine topological connectivity. In the second stage, we subsequently assign directions to these established edges based on the independent directional information.

5 RQ1: Measurement Evaluation

Summary: L2+timestamp+xxgraph algorithm performs best.

6 RQ2: Layer Contribution Analysis

Summary: using QKV layer performs best. K layer contributes most.

7 RQ3: Failure Analysis

Multi-hop Analysis (RQ3): XX

8 RQ4: Integration Evaluation

9 Related Work

9.1 Model Identity Representation

Model watermarking deliberately embedded the triggers into the model to protect its intellectual property. Early mechanisms primarily targeted neural networks to defend against model extraction

Table 3: Lineage Detection Accuracy (%) w.r.t. Different Measurement Metrics, Direction Methods, and Graph Algorithms (RQ1)

		Macro-Accuracy										Micro-Accuracy									
Distances	**Dir.**	**L2**	**Cos.**	**AM**	**AMx**	**SA**	**SR**	**PDF**	**Ghost**	**Matrix**	**HuRef**	**L2**	**Cos.**	**AM**	**AMx**	**SA**	**SR**	**PDF**	**Ghost**	**Matrix**	**HuRef**
MoTHer	K	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	T	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Strict-Directional MDST	K	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	T	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Undirected-MST	K	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	T	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
Atlas Greedy DAG	K	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
	T	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...

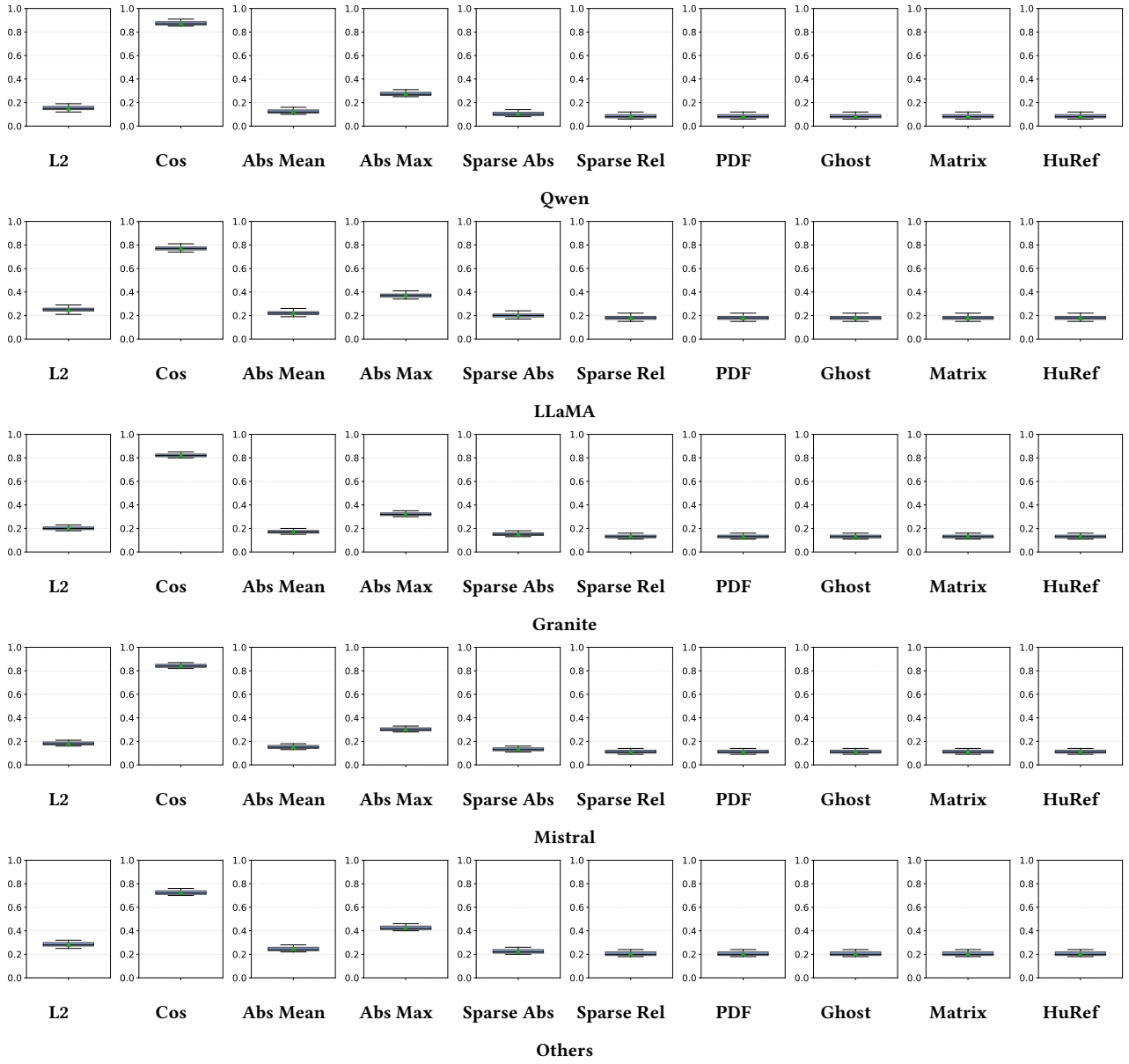


Figure 1: Distance metric distributions for different model families (RQ1).

and surrogate attacks [10, 23, 51, 52, 83, 106, 113]. Subsequent research has focused on embedding invisible and verifiable signatures into generative models [19, 33, 34, 54, 55, 84, 96, 115] to protect AI-generated multimedia content, including images [81], videos [33],

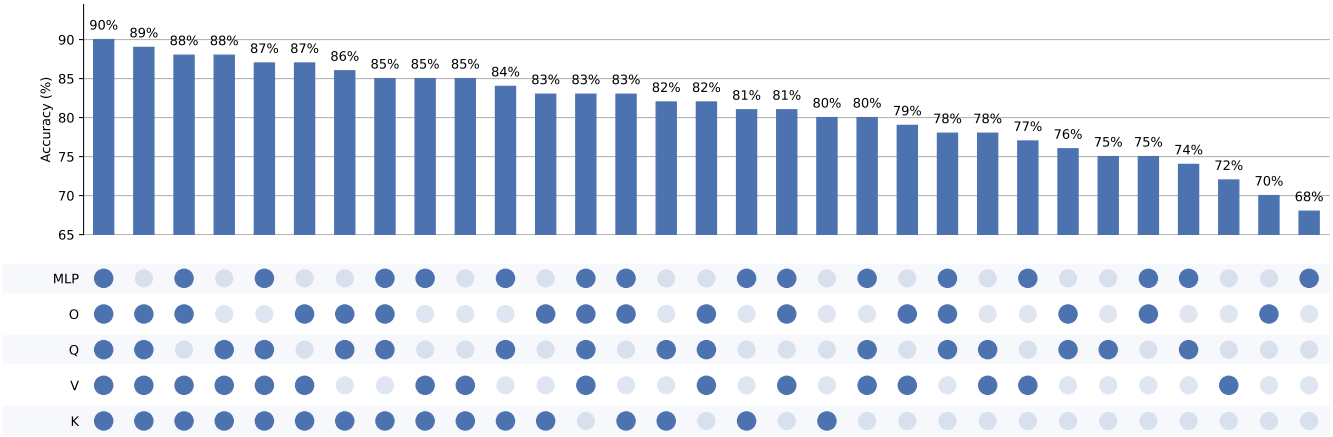


Figure 2: Accuracy of lineage detection with different combinations of transformer block components (RQ2).

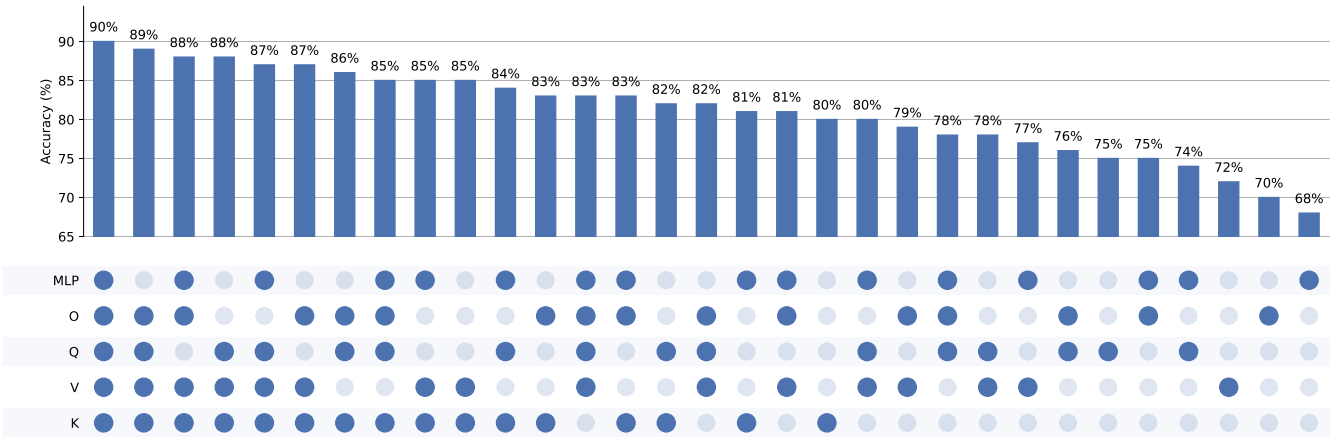


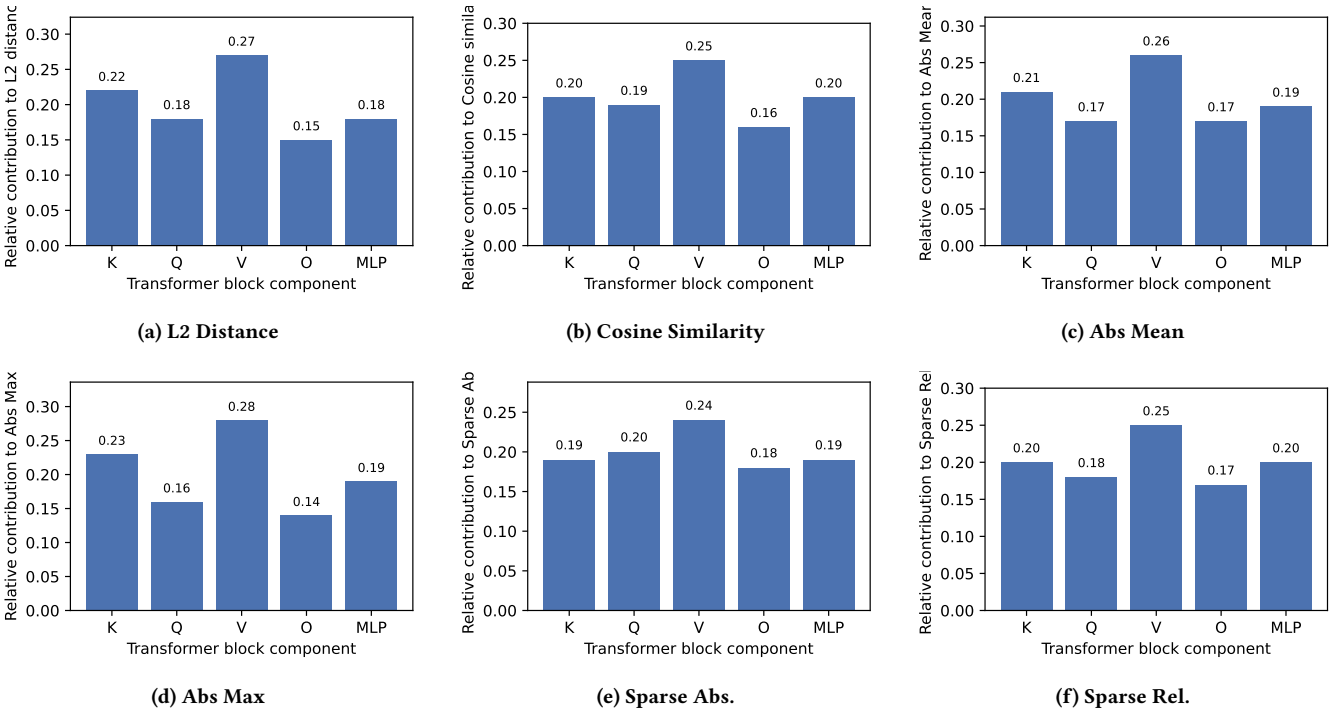
Figure 3: Accuracy2 of lineage detection with different combinations of transformer block components (RQ2).

and audio [40, 115]. Meanwhile, several studies aim to experiment watermark robustness against downstream fine-tuning [82] universal attacks [61], and forgery attacks [61]. Watermarking for LLMs was initially introduced to detect machine-generated text by biasing token selection during decoding [47]. Subsequent work has improved in terms of text quality, embedding capacity, and fine-grained detection capabilities [11, 20, 21, 31, 38, 41, 49, 56, 67, 69, 77, 86, 99, 111]. Meanwhile, adversarial studies have demonstrated evasion via paraphrasing, substitution, and cross-lingual transformations [28, 46, 80, 85]. In addition, recent approaches address challenges such as unforgeability [53] and support multi-user watermarking schemes [15]. Beyond model-level protection, emerging work extends watermarking techniques to data [64], text [112], code [74, 92], and prompts [97].

Model fingerprinting aims to build distinctive identifiers of models, encompassing both black-box behavioral profiling and white-box static analysis. Black-box fingerprinting relies on analyzing a model's dynamic inference behavior in response to carefully crafted inputs [26, 66, 95], including modeling models' outputs as unique vector subspaces [63, 65], functional representations [87], constructed prompts as adversarial suffixes [27, 90, 91]. White-box fingerprinting, in contrast, leverages access to internal model states to derive distinctive model profiles. Dynamic white-box approaches capture runtime characteristics, including feature representations [107], gradient tensors [88], activation differences [89], and token biases [76]. Static white-box fingerprinting captures intrinsic weight characteristics, including statistical distributions [100] (e.g., L_2 distances of model tensors [30]), directionally invariant terms [105], SVD-based spectral signatures [78], and matrix-driven homology proofs [108].

9.2 Model Relation Analysis

Building upon model fingerprinting techniques, recent research has advanced towards reconstructing the model relationships. Using dynamic white-box fingerprints, Wu et al. [88] proposes TensorGuard, which is a gradient-based fingerprinting to detect LLM lineage and classifies model families by analyzing layer-wise gradient responses to input perturbations. Besides, Oyama et al. [63] construct a model map using log-likelihood vectors from over 1,000 models. Wu et al. [87] introduces LLM DNA, a theoretically grounded, low-dimensional representation of model behavior. Using static



(a) L2 Distance (b) Cosine Similarity (c) Abs Mean

(d) Abs Max (e) Sparse Abs. (f) Sparse Rel.

Figure 4: Contribution of different transformer block components (K/Q/V/O/MLP) to various distance metrics (RQ2).

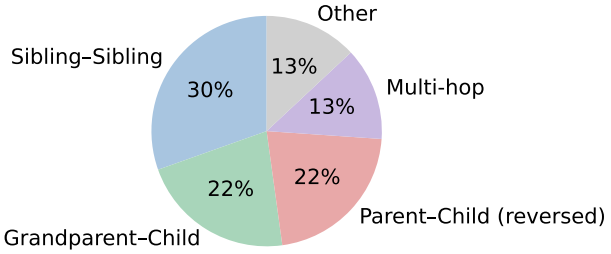


Figure 5: Proportion of different types of erroneous edges in the constructed lineage graph (RQ3).

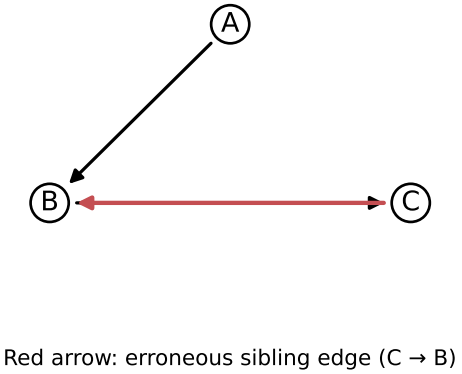


Figure 6: Case study illustrating erroneous edges in the constructed lineage graph (RQ3).

fingerprints, Zhang et al. [107] compares the pairwise feature representations of the two models using centered kernel alignment while remaining robust to model modifications like fine-tuning, pruning, and merging. Zhang et al. [108] proposes to detects and reconstructs weight correspondences using matrix analysis and statistical testing for LLM plagiarism detection without requiring inference. Based on fingerprinting and similarity analysis, recent research has begun to explicitly model LLM lineage. For instance, Yax et al.[98] apply phylogenetic algorithms to LLMs by leveraging output-based similarity to construct evolutionary trees that capture model relationships. Complementing this perspective, Yu et al.[101] propose neural lineage detection methods, including both learning-free and learning-based approaches, to identify parent and ancestral relationships among models. Shifting from behavioral to structural analysis, Horwitz [30] introduces MoTHer, which reconstructs lineage trees by analyzing neural network weights and

inferring parent-child relationships via a directed minimum spanning tree. They further proposes the Model Atlas [29], a unified graph-based framework that captures models, their attributes, and transformation relationships in more comprehensive view of the model ecosystem. We ...

Specifically regarding graph construction, Horwitz et al. [30] build a Model Tree by calculating pairwise weight distances to identify topological connections and utilizing weight kurtosis monotonicity to infer edge directions, ultimately recovering the structure via a minimum directed spanning tree (MDST) algorithm [14, 17]. Expanding this to a global scale, their Model Atlas framework [29] constructs a non-tree Directed Acyclic Graph (DAG) where nodes represent models and edges denote weight transformations. To chart unknown regions of this graph, they employ a greedy algorithm guided by real-world structural priors. This process involves leveraging temporal upload dynamics to determine edge directionality, treating quantized models as leaf nodes, and explicitly accommodating multi-parent merged models to map complex relationships across massive model repositories.

9.3 Software and LLM Supply Chain

Researchers have extensively studied the security and trust challenges in open-source software supply chains. Boughton [8] propose a framework for decomposing and measuring trust in open-source software supply chains by identifying factors that influence the trustworthiness of dependencies. From an adversarial perspective, Ladisa et al.[48] provide a systematic overview of attack techniques targeting open-source software supply chains. Complementary to these studies, Zahan et al. [103] evaluate security metrics and practices in open-source dependencies. Focusing on the JavaScript ecosystem, Zahan [104] analyze npm package metadata and propose weak-link signals to detect potential supply chain risks, while Miller [58] study how the npm ecosystem responds to dependency abandonment and its implications for supply chain sustainability. With the rapid growth of deep learning development, researchers have increasingly focused on the deep learning software supply chain. Tan et al.[75**?**] and Chen et al.[12] conduct exploratory studies on the supply chains of major deep learning frameworks (e.g., PyTorch and TensorFlow) to understand their downstream dependencies, development dynamics, collaboration patterns, and ecosystem differences. From another perspective, some studies focus on faults and failures in the deep learning stack. Huang et al.[36] perform an empirical investigation of dependency-related bugs in the deep learning software stack. Jajal et al.[39] examine interoperability issues of ONNX model converters in deep learning frameworks, highlighting the challenges of cross-framework model portability.

Software composition analysis is widely leveraged to systematically identify software components, supporting tasks such as vulnerability detection [43, 110], license analysis [24], and supply chain risk assessment [62]. The Software Bill of Materials (SBOM) [59, 70, 102] provides a structured description of the internal components of a software system and their dependency relationships. Traditional SCA techniques focus on extracting components from artifacts at different levels. For example, explicit dependencies can be parsed from build configuration files (e.g., package.json and pom.xml) [37, 114]. Static code analysis can be used to identify calls to third-party libraries [**?**]. Binary feature matching can detect repackaged components [93]. With the rise of LLMs, the composition analysis has expanded beyond traditional elements to include models, datasets, frameworks, prompts, and external tools invoked during execution [35, 79]. The concept of SBOM has been extended to the AI domain, such as AI Bill of Materials (AI BOM) [1, 7] and ML Bill of Materials (ML BOM) [4]. At the standardization level, CycloneDX has extended support for ML BOM, and SPDX 3.0 has also begun to discuss and support the modeling of AI-related assets [3]. However, achieving transparent composition analysis for LLM-based systems presents substantial technical challenges. Prior studies have found that such models may memorize training data and leak parts of the training corpus [9, 73]. To infer training data membership, membership inference attacks (MIA) [72] have been adopted. Mattern et al. [57] propose generating contrastive samples that are semantically similar to the target sample to amplify the sample differences. Zhang et al. [109] infer membership by measuring the deviation between the word frequency distribution within a document and that of the overall corpus. Shi et al.b [71] and Li et al. [50] propose the reference-free approaches based on the hypothesis that low-probability tokens tend to appear less frequently in member samples. Nevertheless, Duan et al. [16] found that the boundary between member and non-member samples is inherently fuzzy and continuous in the context of LLMs, leading to limited effectiveness. Besides, Ren et al. [68] studies the evolution of neural networks by constructing a large-scale neural network bill of materials database.

10 Discussion

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